SASA-ESM: Surface-Aware Equivariant Residue Modeling for Protein–Protein Interaction Site Prediction

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Abstract. Protein–protein interaction (PPI) site prediction requires residue-level reasoning over biochemical exposure, sequence context, and three-dimensional geometry. We present SASA-ESM, a reproducible pipeline that constructs weak residue labels from Δ solvent-accessible surface area (SASA), augments each residue with ESM-2 protein language model embeddings and local structural descriptors, and trains an E(n)-equivariant graph neural network (EGNN) for interface classification. The pipeline uses no external graph packages: SASA, ΔSASA labels, residue features, graph construction, EGNN training, cross-chain attention, benchmark manifests, and prediction exports are implemented in the accompanying code. On a 500-complex training corpus (95,005 target-chain residues), a true ESM-2 650M feature set with 1,280-dimensional embeddings and 8 Å residue graph cutoff reaches test F1 = 0.8948, AUROC = 0.9768, and AUPRC = 0.9497. A cross-chain EGNN variant with partner-chain spatial attention obtains comparable test F1 = 0.8938. On the locally analyzable Dset_186 benchmark subset (158 complexes, 39,505 residues), the EGNN model obtains F1 = 0.6448 and AUROC = 0.8868. On PDBtest_315-local (314 complexes, 65,119 residues), the cross-chain EGNN obtains F1 = 0.6816 and AUROC = 0.8983. Ablations over distance cutoffs 6/8/10/12 Å show that 8 Å is the most balanced setting for this corpus. These results support surface-aware equivariant modeling as a compact and reproducible route for PPI site prediction.

Keywords: Protein–protein interaction · Solvent-accessible surface area · Protein language model · ESM-2 · Equivariant graph neural network

1 Introduction

Identifying the amino-acid residues that form a protein–protein interface is a central problem for structural bioinformatics, drug discovery, protein engineering, and functional annotation. Experimental structures provide the most direct evidence of interacting surfaces, but manually deriving residue-level labels is laborious and sensitive to the definition of contact. A practical predictor must combine at least three complementary signals: surface exposure, evolutionary sequence context, and geometric compatibility between nearby residues.

Solvent-accessible surface area (SASA) is a natural physical measure for this task. A residue that is exposed in an isolated chain and becomes buried in a complex is likely to participate in the interface. This observation motivates the common ΔSASA label:

$$\Delta\mathrm{SASA}_i = \mathrm{SASA}_i^{\mathrm{apo}} - \mathrm{SASA}_i^{\mathrm{holo}}, \quad y_i = \mathbb{1}[\Delta\mathrm{SASA}_i \geq \tau].$$

However, SASA alone is not sufficient: exposed non-interface residues are frequent, and interfaces depend on residue type, local secondary geometry, and spatial organization. Recent protein language models such as ESM-2 encode rich sequence context [4], while equivariant graph neural networks can process residue coordinates without discarding the symmetries of three-dimensional space [6].

This work integrates these ingredients into SASA-ESM. The contribution is deliberately engineering-focused and reproducible: rather than assuming external preprocessing, the project implements the full data path from PDB structures to residue-level predictions. The main contributions are:

- a weakly supervised ΔSASA labeling pipeline for two-chain protein complexes;
- a true ESM-2 650M multimodal residue representation combining SASA, 1,280-dimensional protein language model embeddings, backbone/HSE descriptors, hydrophobicity, and Cα coordinates;
- an EGNN and a cross-chain EGNN with partner-chain distance attention for interface prediction;
- benchmark-ready prediction export on Dset_186-local and PDBtest_315-local, plus controlled ablations over graph distance cutoffs.

2 Related Work

Classical interface predictors often use geometric surface descriptors, conservation, physicochemical residue properties, or hand-designed contact features. SASA-based analysis traces back to the rolling-sphere definition of molecular surface accessibility [3], and efficient implementations such as FreeSASA made large-scale surface calculations practical [5]. Graph-based deep learning has since become a common choice for protein interface modeling because residue graphs make local neighborhoods explicit. GraphPPIS is a representative residue graph model for PPI site prediction and introduced widely used benchmark assets such as Test_315 [7].

Protein language models add another axis of information. ESM-2 learns representations from large protein sequence corpora and has been shown to encode structural and functional regularities [4]. In parallel, equivariant networks such as EGNN provide a principled way to use coordinates while respecting translation, rotation, and reflection symmetries [6]. SASA-ESM combines these lines: the label is defined by a physical surface criterion, the node representation includes language-model and local structural features, and the graph model is equivariant.

3 Method

3.1 Data Construction and Labeling

The corpus is built from two-chain protein assemblies downloaded from the RCSB Protein Data Bank [1]. Each complex is parsed into atoms, residues, chains, and Cα residue coordinates. We retain target and partner chains with at least 30 residues to avoid degenerate small fragments.

For each target-chain residue, SASA is calculated twice. The apo value is computed on the target chain alone. The holo value is computed after placing the target chain in the presence of the partner chain. A residue is labeled as interface when ΔSASA ≥ 2.0 Å^2. This threshold is used throughout the reported experiments. Importantly, ΔSASA is only used to define labels; it is not used as an input feature.

3.2 Residue Representation

Each residue node is represented by:

$$x_i = [\mathrm{SASA}_i^{\mathrm{apo}}, \mathrm{SASA}_i^{\mathrm{holo}}, e_i, s_i],$$

where $e_i \in \mathbb{R}^{1280}$ is the ESM-2 650M residue embedding and s_i contains backbone dihedral encodings $(\sin\phi, \cos\phi, \sin\psi, \cos\psi)$, half sphere exposure counts, and hydrophobicity. The resulting true-650M feature set contains 1,289 scalar input features per residue. The code validates this dimensionality to avoid the common failure mode of treating smaller 320-dimensional ESM features as 650M features.

3.3 Equivariant Residue Graph

For each target chain, a residue graph $G = (V, E)$ is built from Cα coordinates. Residues i and j are connected when $\|p_i - p_j\|_2 \leq r$, where r is the distance cutoff. We evaluate $r \in \{6, 8, 10, 12\}$ Å.

The EGNN updates node states and coordinates through messages

$$m_{ij} = \phi_e(h_i, h_j, \|p_i - p_j\|_2/10),$$

with residual feature updates and bounded coordinate weights. The implementation uses unit direction vectors and normalized distances for numerical stability. After three EGNN layers, an MLP classifier outputs an interface logit for every residue.

3.4 Cross-Chain Spatial Attention

The cross-chain variant augments the target-chain EGNN output with partner-chain Cα positions. For target residue i and partner residue j, attention is computed as

$$\alpha_{ij} = \mathrm{softmax}_j(-\|p_i - q_j\|_2/\sigma),$$

PDB complex → apo/holo SASA → ΔSASA labels → ESM-2 650M residue embeddings → SASA + ESM + structural node features → EGNN / cross-chain EGNN → residue interface probabilities

Fig. 1. SASA-ESM processing pipeline. The same manifest drives label generation, embedding extraction, multimodal dataset construction, training, and benchmark prediction export.

Table 1. Datasets used in the current package.

Dataset	Complexes	Residues	Role
Main two-chain corpus	500	95,005	Train/validation/test split
Dset_186-local	158	39,505	External benchmark prediction
PDBtest_315-local	314	65,119	External benchmark prediction

where σ is learnable. Partner coordinates are projected into hidden vectors and combined with target hidden states before classification. This lightweight mechanism lets the model access the spatial layout of the interacting chain without constructing a full bipartite atom graph.

4 Experimental Setup

4.1 Datasets

Table 1 summarizes the datasets used in this manuscript. The main corpus contains 500 two-chain complexes and 95,005 labeled target-chain residues. The Dset_186 evaluation uses the locally analyzable subset produced by the benchmark pipeline after structure availability and chain-quality filtering: 158 complexes and 39,505 residues. PDBtest_315-local uses 314 analyzable complexes and 65,119 residues after the same chain-quality rules. We use the *-local* suffix because both datasets are produced by this repository's reproducible preprocessing protocol.

4.2 Training Protocol

Models are trained with Adam, learning rate 10^{-3}, weight decay 10^{-4}, hidden dimension 128, dropout 0.4, three EGNN layers, gradient clipping at 1.0, and early stopping based on validation F1. Complexes are split by complex identity into 70/15/15 train/validation/test partitions with seed 42. Metrics include accuracy, precision, recall, F1, AUROC, and AUPRC. AUPRC is reported because interface labels are class-imbalanced and precision–recall curves are more informative under skewed class priors [2].

Table 2. Held-out test performance on the 500-complex corpus.

Model	Cutoff	Acc.	Prec.	Rec.	F1	AUROC / AUPRC
EGNN	8 Å	0.9590	0.9023	0.8874	**0.8948**	0.9768 / **0.9497**
Cross-chain EGNN	8 Å	0.9584	0.8976	0.8899	0.8938	0.9755 / 0.9485

Table 3. EGNN distance cutoff ablation on the held-out test split.

Cutoff	Acc.	Prec.	Rec.	F1	AUROC / AUPRC
6 Å	0.9574	0.9027	0.8781	0.8902	0.9750 / 0.9420
8 Å	**0.9590**	0.9023	**0.8874**	**0.8948**	**0.9768** / **0.9497**
10 Å	0.9577	0.9040	0.8777	0.8907	0.9758 / 0.9488
12 Å	0.9584	**0.9188**	0.8645	0.8908	0.9727 / 0.9449

5 Results

5.1 Main Corpus

Table 2 reports the strongest main-corpus configurations. EGNN with the true ESM-2 650M feature set and 8 Å graph cutoff obtains the best test F1 (0.8948) and best AUPRC (0.9497). The cross-chain EGNN performs similarly, with slightly higher recall but lower precision on the held-out split.

5.2 Distance Cutoff Ablation

The graph cutoff controls the amount of geometric context available to each residue. Table 3 shows that 8 Å yields the best F1 and AUPRC, while larger cutoffs do not improve the held-out test set. This suggests that the most useful information is local but not extremely short-range; a 6 Å graph may miss relevant neighbors, whereas 10–12 Å adds edges with weaker interface specificity.

5.3 Dset_186-local Prediction

Table 4 reports prediction metrics on Dset_186-local using the 8 Å models trained on the main corpus. Both models reach similar F1 values. The EGNN has higher precision and AUPRC, while the cross-chain EGNN has higher recall and AUROC. The drop from the internal held-out split to the benchmark subset is expected because the benchmark includes unseen structures and was produced through an independent manifest path.

5.4 PDBtest_315-local Prediction

Table 5 reports the completed PDBtest_315-local run. All 315 official chain-aware entries obtained structures from RCSB; 314 passed the current chain-quality rules. ESM-2 650M embeddings and both model predictions were produced on one NVIDIA GeForce RTX 5060 Laptop GPU. The cross-chain EGNN

Table 4. External prediction on Dset_186-local (158 complexes, 39,505 residues).

Model	Acc.	Prec.	Rec.	F1	AUROC	AUPRC
EGNN	**0.9184**	**0.6690**	0.6223	**0.6448**	0.8868	**0.6770**
Cross-chain EGNN	0.9113	0.6167	**0.6735**	0.6439	**0.8891**	0.5766

Table 5. External prediction on PDBtest_315-local (314 complexes, 65,119 residues).

Model	Acc.	Prec.	Rec.	F1	AUROC	AUPRC
EGNN	**0.9136**	**0.7360**	0.6252	0.6761	0.8946	**0.7408**
Cross-chain EGNN	0.9107	0.7014	**0.6629**	**0.6816**	**0.8983**	0.6906

improves recall, F1, and AUROC, while the target-only EGNN retains the higher precision and AUPRC.

6 Discussion

The experiments show that a physically grounded label source and a language-model representation are complementary. SASA provides the supervision signal and two direct surface features; ESM-2 contributes sequence and evolutionary context; structural descriptors and coordinates let the EGNN model local geometry. The true-650M correction is important: the final files contain 1,280 ESM dimensions and 1,289 total features, whereas older 320-dimensional embeddings would correspond to a different ESM-2 scale and cannot be compared as 650M experiments.

Cross-chain attention did not clearly outperform the target-only EGNN on the main split. This is still informative. The target chain already contains strong interface cues in ΔSASA-derived labels, SASA features, and local coordinates. The partner-chain attention is beneficial on the harder PDBtest_315-local benchmark for recall, F1, and AUROC, but it does not consistently improve precision or AUPRC. A future training setup should explicitly sample non-interacting partner candidates.

7 Limitations and Reproducibility

This manuscript reports only experiments completed in the packaged project state. PDBtest_315-local preprocessing, ESM-2 extraction, and prediction are complete. The collector can be configured for 1,000+ complexes and the extractor supports larger ESM-2 models such as 3B, but those scale-up experiments should be run and reported separately with storage, runtime, and hardware details.

All reported numeric tables are backed by CSV files included with this submission package under `data/`. The implementation exports per-residue predictions for benchmark analysis and stores model checkpoints with feature columns, normalization statistics, cutoff, hidden size, and metric metadata.

8 Conclusion

SASA-ESM provides a compact, end-to-end implementation for PPI site prediction from protein complex structures. On a 500-complex corpus, true ESM-2 650M features and EGNN geometry reach F1 near 0.895 and AUROC near 0.977. Dset_186-local and PDBtest_315-local results demonstrate that the model can be exported to independent benchmark manifests. The current package is suitable as an ICONIP-style LNCS submission draft and a foundation for planned 1,000+ complex and larger ESM model extensions.

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